

Partha Ghosh

67, Swamibag Lane, Dhaka, Bangladesh

+8801912896129 | parthaghoshub361@gmail.com | github.com/ppaarth | linkedin.com/in/ppaarth

Education

Shahjalal University of Science and Technology

BSc in Computer Science and Engineering (Graduated)

- CGPA 3.35 out of 4.00

Sylhet, Bangladesh

February 2018 - March 2023

Work Experience

CodeLab FZC

Software Engineer

- Build a 2D object detection game using the given hint by unity for both android and ios.
- Build a 2D japanese typing game where a player need to complete multiple sentence between the given time by unity.
- Technologies:** Unity, C#

June 2023 - Present

Skills

Programming

Python(Django), C/C++, C#(Unity), Java(Android), Machine Learning(Pytorch), SQL, Javascript(Node.js, React.js), HTML/CSS, PHP(Laravel)

Miscellaneous

Linux, Shell (Bash), LaTeX (Overleaf), Microsoft Office, Firebase, Git

Problem Solving

Solved 600+ problems in different online judge including

Codeforces([partha361](#)), Leetcode([ppaarth](#)), lightoj([parthaghoshu](#)), Vjudge Solves([2017331039](#)), Stopstalk([ppaarth](#))

Personal Projects

SUST Student Management System

A web application which manages all the student , faculty information and results of a student

- University Administration can add, update or delete any student and faculty member and also can add departments and courses and assign faculty members to the appropriate courses. Faculty members can add marks for all students who are registered for the course and students can see their results via this application.
- Django, Python, HTML, CSS, Bootstrap
- Github:** [github.com/ppaarth/SUST Student Management System](#)

Nov 2021 - Dec 2021

Food Book

An android application which is a recipe app

- Anyone can upload any recipe with a description, making time and needed ingredients. Anyone can follow another user, see other's recipes and save them to his/her own profile. Users can comment on it and also can give a rating on the recipe. User also get a notification if anyone follows the user or give comment on the recipe.
- Android, Java, XML, Firebase
- Github:** [github.com/ppaarth/FOOD-BOOK](#)

Aug 2019 - Oct 2019

Muktokathan A Chat app

An android app for chatting

- A chat app by which a user can add friends and can chat with friends. User can search any friend or other users.
- Android, Java, XML, Firebase
- Github:** [github.com/ppaarth/Muktokathan A Chat app](#)

Nov 2021 - Dec 2021

Automatic Attendance System Using Face Recognition

A web application using streamlit which can recognize the face and mark attendance

- We use face recognition to recognize the face which is in front of the webcam and mark the attendees who are attending. We deploy the method in a web app using streamlit where we use the webcam to detect and recognize the face. We collected 5 pictures of every person of our classmates to recognize the face.
- Python, Computer vision, Streamlit
- Github:** [github.com/ppaarth/Automatic-Attendance-System-Using-Face-Recognition](#)

Aug 2022 - Sep 2022

Shoesly an E-commerce Website

A shoe selling e-commerce website

- User, Supplier, Admin and Bank API are the 4 routes in our application that can filter the products with the different price ranges and categories and can confirm the order with the successful transaction via bank. Admin transfers the corresponding charges via the bank to the supplier. Supplier can confirm the order and supply the order to its appropriate customer.
- Node.js, React.js, MongoDB, Express.js
- Github:** [github.com/ppaarth/Shoesly-a-web-application](#)

Jun 2022 - Aug 2022

Automatic Speech Recognition

Mar 2021 - Apr 2021

Machine Learning Project

- Used SHRUTI bangla corpus for automatic speech recognition. Used Deep Speech2 model to train dataset for speech recognition.
- Python, Deep Learning, RNN

Research Experience

Development of a Large Corpus of Spontaneous Speech for Improved Speech Recognition in Bangladeshi Bangla

2022 - Current

For the first time in bangla, we built 200 hours of large corpus of spontaneous speech in Bangladeshi Bangla language to improve the ASR system's accuracy. We fine-tuned multilingual wav2vec 2.0 architecture on our spontaneous speech corpus. We made comparison between reading and spontaneous speech corpus.

Investigating the effect of domain selection on automatic speech recognition performance: a case study on Bangladeshi Bangla

2022 - 2023

- We demonstrate the significance of domain selection by assessing a state-of-the-art Bangla ASR model on a novel multi-domain Bangladeshi Bangla ASR evaluation benchmark— BanSpeech, which contains 7.2 hours of speech and 9802 utterances from 19 distinct domains.
- Experimental evaluation reveals the ASR model on SUBAK.KO faces difficulty recognizing speech from domains with mostly spontaneous speech and has a high number of out-of-vocabulary words.
- **Paper Link:** arxiv.org/abs/2210.12921

Development of a Chakma Corpus for TTS System

2023 - Current

We are building a Chakma language corpus for TTS system and collecting data from a chakma speaker.

Achievements

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|------|---|
| 2023 | 1st Runner up in AI For Bangla 2.0 |
| 2022 | Participated on Robi Datathon 2.0 on kaggle(parthaaghosh), DLSprint |

Extra-Curricular Activities

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| 2022 | General Secretary in CSE Society, SUST |
| 2020 | Assistant General Secretary in CSE Society, SUST |
| 2019 | Executive Member in CSE Society, SUST |
| 2022 | Joint Secretary in SUST ACM Student Chapter |

Reference

Dr. Mohammad Shahidur Rahman

Professor

Department of CSE, SUST

Phone: 01914930807

Email: rahmanms@sust.edu